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1. Introduction

Let M be a smooth compact connected manifold, orientable and without boundary, of dimension m. Denote by $\mathcal{C}^\infty(M)$ the Fréchet space of smooth functions on M and by $\mathcal{V}^\infty(M)$ the Fréchet space of smooth top degree forms on M. Denote by $\mathcal{C}^{-\infty}(M)$ the topological dual of $\mathcal{V}^\infty(M)$, whose elements will be called generalized functions on M, and by $\mathcal{V}^{-\infty}(M)$ the topological dual of $\mathcal{C}^\infty(M)$, whose elements will be called generalized measures on M. Note that measures belong to $\mathcal{V}^{-\infty}(M)$.

Let now X be a smooth vector field on M and denote by $\mathcal{L}_X$ the Lie derivative along X.

1.1. Definition (Globally hypoelliptic/Cohomology-free vector field)**.** The vector field X is called *globally hypoelliptic* if, whenever u is a generalized function on M such that $\mathcal{L}_X u$ is smooth, we have that u is smooth.

The vector field X is called *cohomology-free* if the map

$$\mathcal{L}_X : \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$$

has closed range and its cokernel is one-dimensional.

A vector field is globally hypoelliptic if and only if it is cohomology-free, by [For08, Thm. 3.10] (extending the proof on the torus by Chen–Chi [CC00, Thm. 2.2], see also [RHRH06, §4]). We will refer to a vector field satisfying either of the equivalent assumptions in Definition 1.1 as a KGW-vector field, for [Kat01, Conj. 3.6] (who called them cohomology-free) and Greenfield–Wallach [GW73, Problem 2] who called them globally hypoelliptic. In each of these works, it was conjectured that the only KGW vector fields are linear Diophantine ones on tori.

KGW vector fields satisfy a number of known properties, including the following:

(i) The vector space of invariant generalized measures is one-dimensional, spanned by a smooth positive fully supported volume form on M normalized to $\langle \mu, 1 \rangle = 1$.
(ii) On functions with zero average for μ, denoted $\mathcal{C}^\infty_0(M)$, the operator $\mathcal{L}_X$ is an isomorphism of Fréchet spaces.

Our main result is:

1.2. Theorem (KGW has zero entropy)**.** *The flow generated by a KGW-vector field has zero topological entropy.*

Our proof is based on several ingredients, of possibly independent interest, which we now describe.

The first is a general property of dynamical systems with positive entropy on manifolds; as will be clear from the proof, the result can be generalized

to other spaces with polynomial growth of the number of balls of a given radius needed to cover the space.

1.3. Lemma (Exponential recurrence)**.** *Let $g_t\colon M \to M$ be a smooth flow on a compact manifold, with positive topological entropy and no fixed points. There exist $c, \varepsilon > 0$, a sequence of points $x_i, y_i \in M$, and times $T_i \to +\infty$, satisfying:*

(i) $d(x_i, y_i) \leq e^{-\varepsilon T_i}$

(ii) $d\left(g_{T_i} x_i, g_{T_i} y_i\right) \leq e^{-\varepsilon T_i}$

(iii) *For any continuous strictly increasing $\rho\colon [0, T_i] \to [0, T_i]$ with $\rho(0) = 0, \rho(T_i) = T_i$, there exists $t_{i,\rho} \in [0, T_i]$ such that $d\left(g_{\rho(t_{i,\rho})} x_i, g_{t_{i,\rho}} y_i\right) \geq c$.*

The second result we will need applies to ergodic integrals of the flow g_t generated by a KGW vector field X. Denote by $\delta_{[x, g_t x]}$ the measure on M supported on the corresponding orbit, defined explicitly on functions by

$$\left\langle \delta_{[x, g_t x]}, f \right\rangle := \int_0^t f(g_s x) ds.$$

We then have:

1.4. Lemma (Endpoint distributions)**.** *Suppose X is a KGW vector field. Then there exists a family of generalized measures $\xi_x \in \mathcal{V}^{-\infty}(M)$, defined for every $x \in M$, with the following properties:*

(i) *We have the identity of generalized measures:*

$$\delta_{[x, g_t x]} = t\mu + (\xi_{g_t x} - \xi_x) \tag{1.5}$$

(ii) *We have the Lipschitz bound:*

$$\|\xi_x - \xi_y\|_{\mathcal{V}^{-k_0}} \lesssim d(x, y) \text{ and } \quad \|\xi_x\|_{\mathcal{V}^{-k_0}} \lesssim 1 \quad \forall x, y \in M \tag{1.6}$$

for some $k_0 \in \mathbb{N}$, where $\mathcal{V}^{-k_0}$ denotes the norm on generalized measures obtained by testing against $\mathcal{C}^{k_0}$-functions.

The last result has an elementary but useful consequence:

1.7. Corollary (Diophantine properties)**.** *Suppose X is a KGW vector field, and $\dim M \geq 2$. Then there exist $C, c, A, a > 0$ such that, for any $x \in M$ and $T \geq 1$ we have:*

$$d(x, g_T x) \geq \frac{c}{T^A} \tag{1.8}$$

and for any $y \in M$ there exists $t \in [0, T]$ such that

$$d(y, g_t x) \leq \frac{C}{T^a}. \tag{1.9}$$

Note that the Diophantine properties in this corollary are the conjectured ones in the case of a linear flow on the torus. At the same time, the horocycle flow on a compact quotient of $\mathrm{SL}_2(\mathbb{R})$ satisfies the conclusions of this corollary.

The proofs of the above results are in the next section.

1.10. Discrete-time case. We note that the main result Theorem 1.2 generalizes straightforwardly to the discrete-time case. Namely, a diffeomorphism $f\colon M \to M$ is *cohomology-free* if the operator $f^* - \mathrm{id}$ acting on $\mathcal{C}^\infty(M)$ has image a closed subspace of codimension 1. Then such a diffeomorphism satisfies the analogues of all the above stated results. The proofs are essentially the same, with the operator $\mathcal{L}_X$ replaced by $f^* - \mathrm{id}$. Lemma 1.3 becomes in fact easier, as one can appeal to the Brin–Katok local entropy formula [BK83], one does not need to consider reparametrized orbits, and the matching Lemma 2.14 below is straightforward for discrete time dynamics.

2. Proofs

We will use the notation $A \lesssim B$ when there exists a constant $c > 0$ such that $A \leq cB$, where c is allowed to depend on the manifold and vector field and perhaps other a priori fixed choices (e.g. charts, Riemannian metric, etc.). We will also write $A \asymp B$ for $A \lesssim B$ and $B \lesssim A$.

2.1. Proof of Lemma 1.3. We divide the proof into two natural parts.
Step 1: Entropy considerations. Consider the set of reparametrizations:

$$Rp[0,T] = \{\rho\colon [0,T] \to [0,T] \text{ orientation-preserving homeomorphism}\}.$$

Recall that ordinary Bowen balls are defined by

$$B^T(x,r) := \{y \in M\colon d(g_t x, g_t y) < r, \quad \forall t \in [0,T]\}$$

and consider the Bowen balls with "reparametrization":

$$B^T_{rp}(x,r) := \{y \in M\colon \exists \rho \in Rp[0,T] \text{ s.t. } d(g_{\rho(t)}x, g_t y) < r, \forall t \in [0,T]\}$$

We note for future reference that

$$y \in B^T_{rp}(x,r) \iff x \in B^T_{rp}(y,r) \tag{2.2}$$

by considering the inverse parametrization, and similarly:

$$y \in B^T_{rp}(x,r_1) \text{ and } z \in B^T_{rp}(y,r_2) \implies z \in B^T_{rp}(x,r_1+r_2). \tag{2.3}$$

Denote by $N^T_{rp}(r)$ the minimal number of $x_i \in M$ such that $B^T_{rp}(x_i,r)$ cover M, and $N^T(r)$ for the minimal number using ordinary Bowen balls. Set

$$h_{rp}(g_\bullet) := \lim_{r\to 0} \limsup_{T\to+\infty} \frac{1}{T} \log N^T_{rp}(r)$$

and analogously $h_{top}(g_\bullet)$, which is the topological entropy of the flow, which is also the topological entropy of the diffeomorphism g_1.

A result of Thomas [Tho90, Thm. 10] implies that $h_{rp}(g_\bullet) = h_{top}(g_\bullet) > 0$. Indeed, it is clear that the reparametrized Bowen balls are at least as large as ordinary Bowen balls, and the balls used by Thomas in loc. cit. are larger than the $B^T_{rp}(x,r)$ defined above (Thomas uses a larger class of reparametrizations).

Let $E^T(r) \subset M$ denote a set of minimal cardinality such that $B^T_{rp}(x_i, r)$ cover M. It follows that there exists $r > 0$ and a sequence $T_i \to +\infty$ such that we have

$$\left|E^{T_i}(r)\right| = N^{T_i}_{rp}(r) \geq e^{T_i\beta} \quad \text{for } \beta := \tfrac{1}{2}h_{top}(g_\bullet) > 0.$$

Note that $N^T_{rp}(r') \geq N^T_{rp}(r)$ if $r' < r$. We work with T belonging to the sequence T_i just obtained.

Consider now a subset $\hat{E}^T(r/4)$ of $E^T(r/4)$, of maximal cardinality with the property that the balls $B^T_{rp}(\hat{e}, r/4)$ over all $\hat{e} \in \hat{E}^T(r/4)$ are disjoint. Then we claim that the union of all the balls $B^T_{rp}(\hat{e}, r)$ over all $\hat{e} \in \hat{E}^T(r/4)$ cover M and hence

$$\left|\hat{E}^T(r/4)\right| \geq \left|E^T(r)\right| \geq e^{T\beta}. \tag{2.4}$$

Indeed, for any $z \in M$ there exists $y \in E^T(r/4)$ such that $z \in B^T_{rp}(y, r/4)$. If $y \in \hat{E}^T(r/4)$ we are done, if not y cannot be added in so there exists $x \in \hat{E}^T(r/4)$ such that $B^T_{rp}(x, r/4) \cap B^T_{rp}(y, r/4) \neq \emptyset$. It follows from Eqn. (2.2) and Eqn. (2.3) that $z \in B^T_{rp}(x, r)$.

Step 2: Polynomial pigeonhole. Since M is a compact manifold of dimension m, there exists a constant $C \geq 1$ such that for any $\delta > 0$, there exists a set $P_\delta \subset M$ of cardinality at most $C\delta^{-m}$ such that every point in M is at (ordinary) distance at most δ to some element of P_δ. Set $\delta = \delta_T := e^{-T\frac{\beta}{4m}}$, so T remains the only free parameter. Set also $E^T := \hat{E}^T(r/4)$ satisfying the exponential lower bound in Eqn. (2.4).

Choose now maps $\alpha, \omega \colon E^T \to P_\delta$ such that

$$d(s, \alpha(s)) \leq \delta \quad \forall s \in E^T$$
$$d(g_T s, \omega(s)) \leq \delta \quad \forall s \in E^T$$

which is possible by the properties of P_δ. Now the cardinality of $P_\delta \times P_\delta$ is at most $C^2 e^{T\beta/2}$, while the cardinality of E^T is at least $e^{T\beta}$ by Eqn. (2.4). Assuming $T \geq T_0$ for T_0 sufficiently large, the map $\alpha \times \omega \colon E^T \to P_\delta \times P_\delta$ cannot be injective. Thus, there exist distinct $s_1, s_2 \in E^T$ with the following properties:

(i) $d(s_1, s_2) \leq 2e^{-T\frac{\beta}{4m}}$ since $\alpha(s_1) = \alpha(s_2)$.
(ii) $d(g_T s_1, g_T s_2) \leq 2e^{-T\frac{\beta}{4m}}$ since $\omega(s_1) = \omega(s_2)$.
(iii) The balls $B^T_{rp}(s_1, r/4)$ and $B^T_{rp}(s_2, r/4)$ are disjoint, in particular $s_2 \notin B^T_{rp}(s_1, r/4)$.

Passing now to the sequence $T = T_i \to +\infty$ completes the proof of Lemma 1.3. □

2.5. Proof of Lemma 1.4. We know that $\mathcal{L}_X : \mathcal{C}^\infty_0(M) \to \mathcal{C}^\infty_0(M)$ is an isomorphism of Fréchet spaces. Denote by $\mathcal{G}$ its inverse, extended to all smooth

functions by the condition $\mathcal{G}(1) := 1$, which is a continuous isomorphism of $\mathcal{C}^\infty(M)$. Therefore, its dual is a continuous isomorphism

$$\mathcal{G}^t \colon \mathcal{V}^{-\infty}(M) \to \mathcal{V}^{-\infty}(M).$$

Note that we can decompose any generalized measure ξ as $\xi = \xi_0 + \langle \xi, 1\rangle \, \mu$ where $\langle \xi_0, 1\rangle = 0$. Taking into account the normalization $\mathcal{G}(1) = 1$, we have that $\mathcal{G}^t(\mu) = \mu$ and furthermore

$$\mathcal{G}^t(\xi) = \mathcal{G}^t(\xi_0) + \langle \xi, 1\rangle \, \mu$$

with the property that $\langle \mathcal{G}^t(\xi_0), 1\rangle = 0$.

The continuity of $\mathcal{G}$ implies that there exists $k_0 \in \mathbb{N}$ such that

$$\mathcal{G} \colon \mathcal{C}^{k_0}(M) \to \mathcal{C}^1(M) \text{ is bounded.}$$

Therefore

$$\mathcal{G}^t \colon \mathcal{V}^{-1}(M) \to \mathcal{V}^{-k_0}(M) \text{ is bounded.} \tag{2.6}$$

For a point $x \in M$ denote by $\delta_x \in \mathcal{V}^{-\infty}(M)$ the Dirac-delta measure supported at x. Then $\delta_x \in \mathcal{V}^0(M)$, i.e. it defines a continuous linear functional on continuous functions, and in the weaker $\mathcal{V}^{-1}$-norm we have:

$$\|\delta_x - \delta_y\|_{\mathcal{V}^{-1}} \leq d(x, y)$$

which follows from

$$|\langle \delta_x - \delta_y, \phi\rangle| \leq d(x, y) \, \|\phi\|_{\mathcal{C}^1}$$

for any once-differentiable ϕ. Set

$$\xi_x := \mathcal{G}^t(\delta_x) \text{ so that } \mathcal{L}_X(\xi_x) = \mu - \delta_x \tag{2.7}$$

More explicitly, for a smooth test function ϕ we have $\langle \xi_x, \phi\rangle = u(x)$ where u is the unique solution to $\mathcal{L}_X u = \phi - \int \phi d\mu$ with $\int u d\mu = 0$. To justify Eqn. (2.7) note that $\mathcal{G}\mathcal{L}_X\phi = \phi - \langle \mu, \phi\rangle$ for any smooth test function ϕ and therefore

$$\begin{aligned}\left\langle \mathcal{L}_X \mathcal{G}^t \delta_x, \phi \right\rangle &= -\left\langle \mathcal{G}^t \delta_x, \mathcal{L}_X \phi \right\rangle = -\langle \delta_x, \mathcal{G}\mathcal{L}_X\phi\rangle \\ &= -\langle \delta_x, \phi - \langle \mu, \phi\rangle\rangle = \langle \mu - \delta_x, \phi\rangle \, .\end{aligned}$$

Note the estimate

$$\|\xi_x - \xi_y\|_{\mathcal{V}^{-k_0}} = \left\| \mathcal{G}^t(\delta_x - \delta_y) \right\|_{\mathcal{V}^{-k_0}} \lesssim \|\delta_x - \delta_y\|_{\mathcal{V}^{-1}} \leq d(x, y)$$

where the middle inequality follows from Eqn. (2.6), which implies the second bound in Eqn. (1.6) as well.

The claimed identity Eqn. (1.5)

$$\delta_{[x, g_t x]} = t\mu + (\xi_{g_t x} - \xi_x)$$

follows by observing that both sides agree after applying $\mathcal{L}_X$, hence the difference is $\mathcal{L}_X$-invariant so a multiple of μ, and both sides yield t when

paired against the constant function 1, so must agree. Indeed $\mathcal{L}_X(\delta_{[x,g_tx]}) = \delta_x - \delta_{g_tx}$, since

$$\begin{aligned}\left\langle \mathcal{L}_X \delta_{[x,g_tx]}, \phi \right\rangle &= -\left\langle \delta_{[x,g_tx]}, \mathcal{L}_X \phi \right\rangle = -\int_0^t (\mathcal{L}_X \phi)(g_s x) ds \\ &= -[\phi(g_t x) - \phi(x)] = \langle \delta_x - \delta_{g_tx}, \phi \rangle\end{aligned}$$

and $\mathcal{L}_X(\xi_{g_tx} - \xi_x) = \delta_x - \delta_{g_tx}$ by Eqn. (2.7). This completes the proof of Lemma 1.4. □

2.8. Preliminary constructions in charts. For the proofs of the remaining statements, we will exploit the estimates on the generalized measures ξ_x obtained in Lemma 1.4 by testing them against bump functions concentrated at polynomially small scales and centered at appropriately selected points in the manifold. To do so, we need some preliminaries.

Fix finitely many coordinate charts covering M, with images convex sets in $\mathbb{R}^m$. For each point of M choose one of the charts such that the distance (in $\mathbb{R}^m$) from the point to the boundary of the chart is uniformly bounded below by some $r_{ch} > 0$.

We also assume that in these charts, the vector field is "rectified" to be equal to $\frac{\partial}{\partial x_1}$ where x_1 is the first coordinate. This rectification is possible since the vector field has no fixed points, since the only invariant measure is volume. Finally, we will measure distances in these charts using Euclidean distances, which yield estimates for the distance in the manifold up to uniform factors.

Fix a smooth nonnegative bump function ρ on $\mathbb{R}^m$, with support in the ball of radius 1, and equal to 1 on the ball of radius $\frac{1}{2}$. Denote by $\rho_r(x) := \rho(x/r)$ and, for $p \in M$ denote by $\rho_{r,p} \in \mathcal{C}^\infty(M)$ the placement of ρ_r in M, using the chart associated to p, and centered at p. We will only take $r > 0$ satisfying $r \in (0, r_{ch})$. We record the following properties of the functions:

$$\|\rho_{r,p}\|_{\mathcal{C}^{k_0}} \lesssim r^{-k_0} \tag{2.9}$$

$$\langle \mu, \rho_{r,p} \rangle \asymp r^m \tag{2.10}$$

2.11. Proof of Corollary 1.7. To establish Eqn. (1.9), we will consider the bump function $\rho_{r,y}$ for some $y \in M$, and pair it against the distributional identity from Eqn. (1.5):

$$\left\langle \delta_{[x,g_Tx]}, \rho_{r,y} \right\rangle = \langle T\mu + (\xi_{g_Tx} - \xi_x), \rho_{r,y} \rangle \,. \tag{2.12}$$

For any $y \in M$ we have

$$\langle T\mu, \rho_{r,y} \rangle \asymp T r^m \text{ and } |\langle \xi_{g_Tx} - \xi_x, \rho_{r,y} \rangle| \lesssim r^{-k_0}.$$

where we used that $\|\xi_p\|_{\mathcal{V}^{-k_0}} \lesssim 1$ for all $p \in M$. Select now $r := T^{-\beta}$, so the first term is of the order $T^{1-m\beta}$ while the second is bounded above by $T^{\beta k_0}$. Choosing $\beta > 0$ sufficiently small (e.g. $\beta := \frac{1}{2(m+k_0)}$) guarantees that the right-hand side of Eqn. (2.12) is nonzero. Therefore, the orbit segment

$[x, g_T x]$ must pass through the $CT^{-\beta}$-neighborhood of y, for some $C > 0$ sufficiently large but fixed.

To establish Eqn. (1.8), we note that there exists a constant $b > 0$ such that for any orbit segment of length T, there exists some $y \in M$ which is at distance at least $\frac{b}{T^{1/(m-1)}}$ from any point in the orbit. This is valid for any smooth flow: the R-neighborhood of the orbit segment has volume at most $B(R^m + R^{m-1}T)$ for some $B > 0$, by considerations in local charts: cover an orbit of length T by $O(1+T/R)$ balls of radius R. By choosing $R = \frac{b}{T^{1/(m-1)}}$ for sufficiently small $b > 0$, some point in M will not be covered.

Returning to Eqn. (1.5), choose such a $y \in M$ and $r := bT^{-\gamma}$ with $\gamma := \frac{1}{m-1}$. Then $\left\langle \delta_{[x,g_T x]}, \rho_{r,y} \right\rangle = 0$. At the same time

$$| \langle \xi_{g_T x} - \xi_x, \rho_{r,y} \rangle | \lesssim d(x, g_T x) T^{k_0 \gamma}$$

while $\langle T\mu, \rho_{r,y} \rangle \asymp T^{1-\gamma m}$. It then follows that

$$d(x, g_T x) \gtrsim T^{1-\gamma(m+k_0)} = T^{-\frac{k_0+1}{m-1}}$$

and this concludes the proof. □

2.13. Proof of Theorem 1.2. Suppose, by contradiction, that $g_t \colon M \to M$ has positive topological entropy. Note that the case of $\dim M \leq 1$ is immediate. We will derive a contradiction to Lemma 1.3 (iii), so assume its conclusion holds and let x_i, y_i, T_i be the corresponding points and times.

Set $r := bT_i^{-\gamma}$ and select $b > 0$ sufficiently small, $\gamma > 0$ sufficiently large (e.g. $\gamma = A$ from Eqn. (1.8) works), such that the following holds: if for some $t', t'' \in [0, T_i]$ we have $d(g_{t'} x_i, g_{t''} x_i) \leq 10r$ then $|t' - t''| \leq 10r$ (where the distance is the Euclidean one in any of the fixed rectifying coordinate charts). If a t'' outside this interval existed, it would contradict the Diophantine lower bound on recurrence from Eqn. (1.8) for $g_{t'} x_i$. The same property holds for y_i.

We claim that there exists $x' := g_{t'} x_i$ with $t' \in [0, T_i]$ such that the ball of radius r around x' (in any of the fixed coordinate charts and Euclidean metric) is disjoint from the orbit segment $[y_i, g_{T_i} y_i]$. Assuming this first, we derive a contradiction as follows.

We pair the distributions in Lemma 1.4 against the bump function $\rho_{r,x'}$:

$$\left\langle \delta_{[x_i, g_{T_i} x_i]}, \rho_{r,x'} \right\rangle \gtrsim r \text{ and } \left\langle \delta_{[y_i, g_{T_i} y_i]}, \rho_{r,x'} \right\rangle = 0.$$

At the same time, using the estimate in Eqn. (1.6) together with the exponential closeness of x_i, y_i and $g_{T_i} x_i, g_{T_i} y_i$, we find:

$$\left\| \delta_{[x_i, g_{T_i} x_i]} - \delta_{[y_i, g_{T_i} y_i]} \right\|_{\mathcal{V}^{-k_0}} = \left\| \left(\xi_{g_{T_i} x_i} - \xi_{g_{T_i} y_i} \right) - \left(\xi_{x_i} - \xi_{y_i} \right) \right\|_{\mathcal{V}^{-k_0}} \lesssim e^{-\varepsilon T_i}$$

Since $\|\rho_{r,x'}\|_{\mathcal{C}^{k_0}} \lesssim r^{-k_0} \asymp T_i^{k_0\gamma}$ a contradiction follows for T_i large enough:

$$\begin{aligned} bT_i^{-\gamma} = r &\lesssim \left\langle \delta_{[x_i, g_{T_i}x_i]}, \rho_{r,x'} \right\rangle \\ &\leq \left| \left\langle \delta_{[y_i, g_{T_i}y_i]}, \rho_{r,x'} \right\rangle \right| + \left\| \delta_{[x_i, g_{T_i}x_i]} - \delta_{[y_i, g_{T_i}y_i]} \right\|_{\mathcal{V}^{-k_0}} \cdot \|\rho_{r,x'}\|_{\mathcal{C}^{k_0}} \\ &\lesssim e^{-\varepsilon T_i} T_i^{k_0\gamma}. \end{aligned}$$

It remains to show such an $x' := g_{t'}x_i$ exists. Suppose it did not, in other words we have $B(g_{t'}x_i, r)$ intersects the orbit segment $[y_i, g_{T_i}y_i]$ for every $t' \in [0, T_i]$. Now the existence of a reparametrization $\rho\colon [0, T_i] \to [0, T_i]$ such that

$$d(g_{\rho(t)}x_i, g_t y_i) \lesssim r \asymp T_i^{-\gamma}$$

follows from the general purpose Lemma 2.14, and this contradicts Lemma 1.3 (iii). □

2.14. Lemma (Orbit matching)**.** *Suppose X is a smooth vector field with no fixed points on a compact manifold M, with flow g_t. Assume that for two points $x, y \in M$ and time $T > 0$, their orbit segments have the following properties, for some $r > 0$, $C \geq 1$:*

(i) *If $d(g_{t'}x, g_{t''}x) < 10r$ for some $t', t'' \in [0, T]$, then $|t' - t''| < Cr$, and same for y.*

(ii) *The orbit segment $[x, g_T x]$ is contained in the r-neighborhood of $[y, g_T y]$.*

(iii) *We have $d(x, y) < r$ and $d(g_T x, g_T y) < r$.*

Then, provided r is sufficiently small depending on C, there exists an orientation-preserving homeomorphism $\rho\colon [0, T] \to [0, T]$ such that

$$d(g_{\rho(t)}x, g_t y) < C'r$$

for some $C' > 0$ depending only on M, X, C but not on r.

Proof. Fix a smooth hyperplane field $H(x) \subset TM(x)$ inside the tangent bundle of M, transverse to the vector field X, for example $H = X^\perp$ for the background Riemannian metric. Denote by $H(S, R)$ the restriction of the vectors of length strictly less than R in the bundle H to a subset $S \subset M$. Since M is compact, there exist uniform constants $t_H, r_H > 0$ such that the exponential map is a smooth C^2-bounded diffeomorphism between $H((z, g_{t_H}z), r_H)$ and an open subset of M, uniformly in $z \in M$, and where $(z, g_{t_H}z)$ denotes the corresponding open orbit segment.

Fix rectifying coordinate charts as in §2.8. There exist $K_1, K_2 > 0$ such that using the exponential map on $H([x, g_T x], K_1 r)$, for every $t \in [0, T]$ there exists a unique $\tilde{\rho}(t) \in [-K_2 r, T + K_2 r]$ and $v_t \in H(g_t x)$ such that $\exp(v_t) = g_{\tilde{\rho}(t)}y$. Indeed, existence follows from the containment of the orbit in the r-neighborhood (and assuming r is sufficiently small) and uniqueness follows from the no close returns of the orbit plus the local diffeomorphism property of the exponential map. Furthermore, the map $\tilde{\rho}$ is a local diffeomorphism by the implicit function theorem, provided r is sufficiently small. We thus

obtain an orientation-preserving homeomorphism $\tilde{\rho}\colon [0,T] \to [a, T+b]$ where $|a|, |b| \leq K_2 r$ such that

$$d(g_t x, g_{\tilde{\rho}(t)} y) \leq K_2 r \quad \forall t \in [0,T].$$

After an affine rescaling and considering the inverse, the desired ρ follows. □

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